Assignment 2

1. INTRODUCTION TO CONDITIONALS

> **Question 1: What is a conditional statement in programming?**
> A control flow statement that allows a program to execute different blocks of code depending on whether a given condition evaluates to true or false.

> **Question 2: What does an if statement do?**
> It tests a specific condition; if the condition evaluates to true, the block of code inside the if statement is executed.

> **Question 3: What is the purpose of an else block?**
> It provides a fallback or alternative block of code to run when the condition in the preceding if statement evaluates to false.

> **Question 4: When is an else block executed?**
> It executes only when all corresponding if (and else if) conditions in that structure evaluate to false.

> **Question 5: What is a Boolean expression?**
> An expression or condition that evaluates strictly to one of two logical results: true (represented as non-zero or 1) or false (represented as 0).

Question 6: Write the syntax of a simple if statement.

```
if (condition) {
    // code to execute if condition is true
}
```

2. RELATIONAL LOGIC & ARITHMETIC CHECKS

> **Question 7: What operator is commonly used to test equality?**
> The double-equal operator (==).

> **Question 8: What does the modulus (%) operator return?**
> It returns the integer remainder of a division operation between two numbers.

> **Question 9: How can you check if a number is even?**
> Check if its remainder when divided by 2 is exactly zero: if (num % 2 == 0).

> **Question 10: How can you check if a number is odd?**
> Check if its remainder when divided by 2 is not zero: if (num % 2 != 0).

> **Question 11: What is the purpose of else-if?**
> It allows checking multiple distinct conditions sequentially after an initial if statement fails.

> **Question 12: What is a nested if statement?**
> An if statement placed inside the code block of another preceding if or else statement.

> **Question 13: Can an if statement exist without an else?**
>
> Yes. The `else` block is completely optional if no fallback behavior is needed.

> **Question 14: What is the output when a false condition is tested in an if statement without else?**
>
> There is no output or execution from that block; the program simply skips it and proceeds to the next line of code.

> **Question 15: What is the role of braces {} in C/C++ conditionals?**
>
> They group multiple statements into a single composite block. Without them, only the single immediate next line is treated as part of the conditional structure.

> **Question 16: What does > mean?**
>
> The "Greater Than" relational operator.

> **Question 17: What does < mean?**
>
> The "Less Than" relational operator.

> **Question 18: What does >= mean?**
>
> The "Greater Than or Equal To" relational operator.

> **Question 19: What does <= mean?**
>
> The "Less Than or Equal To" relational operator.

> **Question 20: What does != mean?**
>
> The "Not Equal To" relational operator used to check inequality.

3. MULTI-WAY BRANCHING (SWITCH)

> **Question 21: What is a switch statement?**
>
> A multi-way branch statement that evaluates an integer-compatible expression and matches its result directly against a series of defined `case` constants.

> **Question 22: What is a case label in a switch statement?**
>
> A marker representing a constant value that the switch expression is checked against for a match.

> **Question 23: Why is break used in switch?**
>
> To terminate execution inside the switch block immediately once a match is processed, preventing execution from falling through to subsequent cases.

> **Question 24: What happens if break is omitted?**
>
> Execution falls through into the next case's statements sequentially, regardless of whether that case label matches, until a `break` or the end of the switch block is reached.

> **Question 25: What is the default case in switch?**
>
> An optional catch-all block that executes if none of the explicit `case` labels match the evaluated expression (similar to an `else` block).

> **Question 26: Can switch compare strings in C++?**
>
> No. In C++, standard `switch` structures can only evaluate integer-compatible types (like `int`, `char`, `enum`). They do not support `std::string` objects directly.

> **Question 27: Which statement is better for many constant choices: if-else or switch?**
>
> The `switch` statement is better; it is structurally cleaner, easier to maintain, and often optimized by the compiler using jump tables for faster execution.

4. RANGE CHECKS & LOGICAL COMPOSITIONS

Question 28: How do you test whether a value is positive?

Check if it is strictly greater than zero: `if (val > 0).`

Question 29: How do you test whether a value is negative?

Check if it is strictly less than zero: `if (val < 0).`

Question 30: How do you test whether a value is zero?

Check equality with zero: `if (val == 0).`

Question 31: Write a condition to check if age is at least 18.

`if (age >= 18)`

Question 32: Write a condition to check if marks are above 50.

`if (marks > 50)`

Question 33: What is decision making in programming?

The conceptual process of establishing checkpoints in a program to select a specific execution path based on variable values or run-time updates.

Question 34: Can nested if statements have multiple levels?

Yes, you can nest structures across multiple levels deep, though deep nesting can make code less readable.

Question 35: What is logical AND (&&)?

A logical operator that returns `true on`ly if both conditions on its left and right sides evaluate to true.

Question 36: What is logical OR (||)?

A logical operator that returns `true if` at least one of the surrounding conditions evaluates to true.

Question 37: What is logical NOT (!)?

A unary operator that inverts the logical state of an expression; it turns true into false, and false into true.

5. ALGORITHM BUILDING & ADVANCED LOGIC

Question 38: How do you check if a year is divisible by 4?

Check if its modulo remainder is zero: `if (year % 4 == 0).`

Question 39: What is a leap year?

A calendar year containing 366 days instead of 365. In programming, it must be divisible by 4, but century years (ending in 00) must also be divisible by 400 to qualify.

Question 40: What is the first condition for a leap year?

The year must be evenly divisible by 4 `(year % 4 == 0).`

Question 41: How do you find the largest of two numbers?

```
if (num1 > num2) {
    largest = num1;
} else {
    largest = num2;
}
```

Question 42: How do you find the largest of three numbers?

```
if (a >= b && a >= c) {
    largest = a;
} else if (b >= a && b >= c) {
    largest = b;
} else {
    largest = c;
}
```

> **Question 43: What is an if-else-if ladder?**
>
> A control structure that tests a sequence of distinct conditions one after another until one evaluates to true, or falls through to an optional final `else` block.

> **Question 44: What is the difference between if and switch?**
>
> An `if` structure evaluates relational inequalities and complex multi-logical expressions (e.g., `a > 5 && b < 10`). A `switch` evaluates a single variable directly against specific discrete integer or character constants.

> **Question 45: Can switch use ranges directly?**
>
> No, standard C/C++ switch structures cannot evaluate ranges directly (like `case 50...100:` is a non-standard GNU extension). You must evaluate exact constant metrics.

> **Question 46: What is fall-through in switch?**
>
> The behavior where execution continues into the code of subsequent case blocks sequentially when a `break` statement is missing.

> **Question 47: What data type is usually used for conditions?**
>
> The boolean data type (`bool`).

> **Question 48: Why are conditional statements important?**
>
> They allow a program to make smart runtime decisions, process varying data sets correctly, and dynamically react to different user behaviors rather than executing fixed code linearly.

> **Question 49: Give one real-life example of a conditional statement.**
>
> An ATM withdrawal system: **IF** your account balance is greater than or equal to the requested withdrawal amount, dispense the cash; **ELSE**, display an "Insufficient Funds" warning message.

> **Question 50: Name four conditional structures covered.**
>
> The simple `if` statement, the `if-else` statement, the `if-else-if` ladder, and the `switch` statement.